SUMS OF TWO SQUARES AND HEEGNER NORMS WITH FEW PRIME FACTORS, V: A PRIME PLUS A HEEGNER NORM (THE LINNIK PROBLEM FOR THE NINE DISCRIMINANTS OF CLASS NUMBER ONE)

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Abstract. Parts I–IV studied $N = n + (N - n)$ with both summands norms from an imaginary quadratic field of class number one, a sieve problem of dimension 2 whose remainder is a bilinear form in Kloosterman sums with one argument fixed. This part changes the problem to $N = p + N_D(x, y)$, a rational prime plus a norm from the field of discriminant $-D$, $D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}$. For $D = 4$ this is the problem of Hardy and Littlewood solved by Linnik with the dispersion method. We prove, for all nine D by the divisor-switching argument of Friedlander and Iwaniec, the asymptotic formula $\sum_{p<N} r_D(N-p) = \frac{2\pi}{\sqrt{D}}\, \mathfrak{S}_D(N)\, \pi(N)+o(\pi(N))$ with the singular series in closed form, $\mathfrak{S}_D(N) = (1 + c_D(N)) \prod_{\ell \mid N}(1 - \chi_{-D}(\ell)/\ell) \prod_{\ell \nmid N}(1 + \chi_{-D}(\ell)/(\ell(\ell - 1)))$, where the factor at the ramified prime, $1 + c_D(N) = 1 - \chi_{-D}(N)/(D - 1)$ for odd D and 1 for $D = 4, 8$, comes from the second half of the hyperbola; the proof needs the Bombieri–Friedlander–Iwaniec level $\sqrt{N}(\log N)^A$ for the single residue class N, which is proved for fixed residues and which we state as a hypothesis, and for $D = 4$ the theorem is unconditional by Linnik. The sequence $\{N - p\}$ has sieve dimension 1, but the condition "$N - p$ is a norm" is not a sieve condition: the weighted sequence $r_D(N - p)$ has dimension 1 as well, and no level of distribution for it follows from the Bombieri–Vinogradov theorem, because the hyperbola puts the sieve modulus next to a modulus of size $\sqrt{N}$. So neither Chen's $N = p + P_2$ nor Richert's $N = p + P_3$ transfers to norms, and we say precisely what would be needed. Numerically the asymptotic formula holds to 1% for all nine D at $N \le 2 \cdot 10^5$, and for N even the proportion of primes $p < N$ with $N - p$ a norm with at most two prime factors is between 0.11 and 0.34.

1. Introduction and statement of results

1.1. **The problem.** Let $D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}$, let $K = \mathbb{Q}(\sqrt{-D})$, $\chi = \chi_{-D}$ the Kronecker symbol of conductor D, w_D the number of units $(6, 4, 2, \dots)$, and N_D the principal form, $x^2 + y^2$, $x^2 + 2y^2$ or $x^2 + xy + \frac{D+1}{4}y^2$. Since $h(-D) = 1$, the number of representations of $n \ge 1$ by N_D is

$$r_D(n) = w_D \sum_{d \mid n} \chi(d) = w_D\, \lambda(n), \qquad \lambda = 1 * \chi,$$

and $r_D(n) > 0$ exactly when n is a norm, i.e. when every inert prime divides n to an even power. Parts I–IV [4, 5, 6, 7] treated $N = n + (N - n)$ with both summands norms, through the theta series $\theta_D(z)\theta_D(pz)$; the sieve there has dimension 2 and the remainder beyond the level $1/2$ is a bilinear form in Kloosterman sums with the single argument N fixed. Here we consider instead

$$N = p + N_D(x, y), \qquad p \text{ prime}, \quad (x, y) \in \mathbb{Z}^2,$$

for a large integer N, through the sum

$$S_D(N) = \sum_{p<N} r_D(N - p) = w_D \sum_{p<N} \sum_{d \mid N-p} \chi(d).$$

Date: November 2, 2026 (first-stage draft).

For $D=4$ this is the Hardy–Littlewood problem $N=p+x^2+y^2$, which Linnik solved with the dispersion method [14, Chapter 7]; Hooley [10] had obtained the asymptotic under the extended Riemann hypothesis. For the shifted problem $p=x^2+y^2+1$, Bredihin [2] extended Linnik's argument, and Friedlander and Iwaniec [8] gave a short proof from the Bombieri–Friedlander–Iwaniec theorem [1] on primes in arithmetic progressions to moduli $\sqrt{x}(\log x)^A$. We follow the latter route, for all nine D at once and for the fixed-N problem, and we compute the singular series explicitly, including the factor at the ramified prime that the problem $N=p+N_D(x,y)$ has and the problem $p-1=x^2+y^2$ does not.

1.2. **The asymptotic formula.** The input is the following statement, which is the theorem of Bombieri, Friedlander and Iwaniec [1, Theorem] with the residue class $a=N$ in place of a fixed residue class.

Hypothesis 1.1 (H(N))**.** For every $A>0$, uniformly in $X\le N$,

$$\sum_{\substack{d\le \sqrt{N}(\log N)^A\\(d,N)=1}}\Big|\pi(X;d,N)-\frac{\pi(X)}{\varphi(d)}\Big|\ll_A N\Big(\frac{\log\log N}{\log N}\Big)^2.$$

For $X\le N(\log N)^{-3}$ the bound is trivial ($\pi(X;d,N)\le X/d+1$), so the content is in $N(\log N)^{-3}\le X\le N$. The Bombieri–Vinogradov theorem gives the bound with $d\le\sqrt{N}(\log N)^{-B}$ and the saving $(\log N)^{-A}$, uniformly in the residue class; the extension to $\sqrt{N}(\log N)^A$ is proved in [1] for a fixed residue class a, with the implied constant depending on a, and the uniform versions known to us (Maynard [15], Lichtman [13]) either restrict the moduli to those with a factor of a prescribed size or replace the absolute values by well-factorable weights. Hypothesis H(N) as stated is therefore open; it is what the divisor-switching proof needs, and nothing more.

Theorem 1.2. *Let* $c_D(N)=\varphi(D)^{-1}\sum_{a \bmod D,\ (a,D)=1}\chi_{-D}(N-a)$, *so that* $c_D(N)=-\chi_{-D}(N)/(D-1)$ *for odd* D *and* $c_D(N)=0$ *for* $D=4,8$, *and let*

$$\mathfrak{S}_D(N)=\big(1+c_D(N)\big)\prod_{\ell\mid N}\Big(1-\frac{\chi(\ell)}{\ell}\Big)\prod_{\ell\nmid N}\Big(1+\frac{\chi(\ell)}{\ell(\ell-1)}\Big).$$

(i) *If* H(N) *holds, then*

$$S_D(N)=\frac{2\pi}{\sqrt{D}}\,\mathfrak{S}_D(N)\,\pi(N)+O\Big(N\Big(\frac{\log\log N}{\log N}\Big)^2\Big).$$

(ii) *For* $D=4$ *the asymptotic* $S_4(N)\sim\pi\,\mathfrak{S}_4(N)\,\pi(N)$ *holds unconditionally, with the error term* $O(N(\log N)^{-1-\eta})$ *of Linnik's theorem* [14, Chapter 7].

In particular $\mathfrak{S}_D(N)\gg 1$ *for every* N *(every factor is at least* $1-\frac12$ *at* $\ell=2$ *and* $1-\frac{1}{D-1}$ *at the ramified prime, and the product over* $\ell\nmid N$ *converges absolutely), so that, under* H(N), *every large* N *is a prime plus a norm from* K, *with* $\gg N^{1-\varepsilon}$ *representations.*

The constant is the mean value of r_D: $w_DL(1,\chi_{-D})=2\pi h/\sqrt{D}=2\pi/\sqrt{D}$ by the class number formula, and the singular series is the correction at each prime, with the two halves of the hyperbola contributing 1 and $c_D(N)$ respectively (Section 2). The factor $1+c_D(N)$ is the local density at the ramified prime: for $D=3$, $N_3(x,y)=x^2+xy+y^2\equiv(x-y)^2 \pmod 3$ never is 2 mod 3, so for $N\equiv1 \pmod 3$ the primes $p\equiv2 \pmod 3$ are lost ($\frac12$), for $N\equiv2$ none are lost and the primes $p\equiv2$ give $N-p\equiv0$, which has twice the density of norms ($\frac32$), and for $N\equiv0$ the two effects cancel (1). Section 3 verifies the formula to 1% for all nine D and $N\le2\cdot10^5$; without the factor $1+c_D(N)$ the ratios for $D=3$ are 0.50, 1.00, 1.50 according to N mod 3.

Remark 1.3. (a) The dispersion method of Linnik does not need H(N); it proves (ii) with a power of $\log N$ saved, and it is what makes the problem $N = p + x^2 + y^2$ unconditional. We expect it to apply to the forms N_D as it applies to $x^2 + y^2$ (the inputs are the divisor structure of r_D and the equidistribution of the norms in residue classes, both of which the class number one fields share), but we have not carried the dispersion argument out for $D \neq 4$, and we do not claim (ii) for them. (b) The error term in (i) is $o(\pi(N))$ and no better; this is the error term of [8], and it comes from the level $\sqrt{N}(\log N)^A$ with the saving $(\log\log N/\log N)^2$ of [1].

1.3. **The sieve.** The sequence $\mathcal{A} = \{N - p : \ p < N\}$ is the sequence of Chen's theorem: its sieve dimension is 1, its level of distribution is $1/2$ by the Bombieri–Vinogradov theorem, and the weighted sieve of Richert gives $N = p + P_3$, Chen's switching $N = p + P_2$, for every large even N. None of this uses or produces the condition that $N - p$ be a norm. The relevant sequence is the weighted one, $a_m = r_D(m)$ for $m = N - p$, and its level of distribution is Theorem 1.2 in arithmetic progressions on average over the modulus.

Proposition 1.4. (i) *For $\mathcal{A} = \{N - p\}$ and a prime $q \nmid N$, $\#\{p < N : \ q \mid N - p\} = \pi(N)/(q-1) + O(E(N;q))$, so that $\omega(q) = q/(q-1)$, $\omega(q) = 0$ for $q \mid N$, and $\prod_{w \le q < z}(1 - \omega(q)/q)^{-1} = \frac{\log z}{\log w}(1 + O(1/\log w))$: dimension 1. The level of distribution is $1/2$.*

(ii) *For the weighted sequence $a_m = r_D(m)1_{N-m \text{ prime}}$ and a prime $q \nmid DN$, the expected local density is $\omega_D(q) = (1 + \chi(q))\, q/(q-1)$: $2q/(q-1)$ at the split primes, 0 at the inert ones, and $\sum_{q<z} \omega_D(q)/q = \log\log z + O(1)$: dimension 1 on average, with the half-dimensional structure of the inert primes. The quantity whose average over $q \le N^\theta$ a sieve needs is*

$$\sum_{\substack{p<N \\ q \mid N-p}} r_D(N-p) - \frac{\omega_D(q)}{q}\,\frac{2\pi}{\sqrt{D}}\mathfrak{S}_D(N)\pi(N),$$

and by the hyperbola (Section 2) its control at the level N^θ is equivalent to the equidistribution of the primes $p \equiv N \pmod{qb}$ for $q \le N^\theta$, $b \le \sqrt{N}(\log N)^A$, with the weights $\chi(b)$: moduli of size $N^{1/2+\theta}(\log N)^A$ in the single residue class N. No $\theta > 0$ follows from the Bombieri–Vinogradov theorem, and Hypothesis H(N) *gives $\theta = 0$ only.*

(iii) *If the weighted sequence had the level $\theta = 1/2$ (in the form needed by the weighted linear sieve), Richert's weighted sieve would give $N = p + P_3$ with $N - p$ a norm, for every large admissible N; Chen's switching would need, in addition, the level $1/2$ for the sequences $\{N - q_1q_2m\}$ with q_1, q_2 split primes in prescribed ranges, and would give $N = p + P_2$ with $N - p$ a norm. Neither is proved here.*

Admissible N. For $D \equiv 3 \pmod 8$ the prime 2 is inert, so an even value $N - p$ of N_D is divisible by 4, and $\Omega(N - p) \le 2$ forces $N - p = 4$: the statement "$N = p + P_2$ with $N - p$ a norm" can only hold for even N when $D \in \{3, 11, 19, 43, 67, 163\}$ (the data of Section 3 show the count 0 at every odd N for these D). For $D = 7$ (2 split) and $D = 4, 8$ (2 ramified) there is no parity condition.

The content of Proposition 1.4 is negative: the change of problem removes one of the two obstacles of Parts I–IV, the dimension 2 (the prime side contributes nothing to the dimension), but not the other. The fixed argument N of the Kloosterman sums of Part III reappears as the fixed residue class N of the primes, at moduli qb with $b \asymp \sqrt{N}$, and this is the same square-root barrier in a different guise. What the change of problem does give, at the level $\sqrt{N}(\log N)^A$ of the single class N, is the asymptotic formula of Theorem 1.2, with every local factor explicit; and for $D = 4$ Linnik's theorem gives that unconditionally.

2. Proof of Theorem 1.2

2.1. The singular series.

Proposition 2.1. *For every $N \geq 1$, $\sum_{(d,N)=1} \chi(d)/\varphi(d) = L(1,\chi)\,\mathfrak{E}_D(N)$ with*

$$\mathfrak{E}_D(N) = \prod_{\ell \mid N}\Big(1 - \frac{\chi(\ell)}{\ell}\Big)\prod_{\ell \nmid N}\Big(1 + \frac{\chi(\ell)}{\ell(\ell-1)}\Big),$$

and $c_D(N) = \varphi(D)^{-1}\sum_{(a,D)=1}\chi(N-a)$ equals $-\chi(N)/(D-1)$ for odd D and 0 for $D = 4, 8$.

Proof. The sum is multiplicative in d; at a prime $\ell \nmid N$ the local factor is $1+\sum_{j\geq 1}\chi(\ell)^j/\varphi(\ell^j) = 1 + \frac{\chi(\ell)}{\ell-1}\big(1-\frac{\chi(\ell)}{\ell}\big)^{-1}$, and $\big(1+\frac{\chi(\ell)\ell}{(\ell-1)(\ell-\chi(\ell))}\big)\big(1-\frac{\chi(\ell)}{\ell}\big) = \frac{\ell^2-\ell+\chi(\ell)}{\ell(\ell-1)} = 1 + \frac{\chi(\ell)}{\ell(\ell-1)}$; at $\ell \mid N$ the factor is $1 = (1-\chi(\ell)/\ell)\cdot(1-\chi(\ell)/\ell)^{-1}$. Multiplying out gives $L(1,\chi)\mathfrak{E}_D(N)$. For odd D (a prime), $\sum_{a=1}^{D-1}\chi(N-a) = \sum_{b\not\equiv N}\chi(b) = -\chi(N)$ since $\sum_{b \bmod D}\chi(b) = 0$. For $D = 4$, $\chi_{-4}(N-1)+\chi_{-4}(N-3) = \chi_{-4}(N-1)+\chi_{-4}(N+1) = 0$ because $N-1 \equiv -(N+1) \pmod 4$ and $\chi_{-4}(-1) = -1$. For $D = 8$ and N even, $\{N-a:\ a \text{ odd}\}$ is the set of odd residues modulo 8, on which χ_{-8} sums to 0; for N odd every $N-a$ is even and $\chi_{-8}(N-a) = 0$. □

2.2. Divisor switching.

For $m \geq 1$ write $m = D^k m'$ with $D \nmid m'$. Since $\chi(D^j) = 0$ for $j \geq 1$, $\lambda(m) = \lambda(m')$, and for $(e,D) = 1$, $\chi(m'/e) = \chi(m')\chi(e)$ whenever $e \mid m'$ (χ is real and completely multiplicative). For $y > 0$, splitting the divisors of m' at y and writing the large ones as m'/e,

$$\lambda(m) = \sum_{\substack{d\mid m'\\ d\leq y}}\chi(d) + \chi(m')\sum_{\substack{e\mid m'\\ e<m'/y}}\chi(e) = \sum_{\substack{d\mid m\\ d\leq y}}\chi(d) + \chi(m')\sum_{\substack{e\mid m'\\ e<m'/y}}\chi(e), \tag{1}$$

the first sum being the same over m and m' because $\chi(d) = 0$ unless $(d,D) = 1$. Let $w \geq 0$ be smooth, supported in $[1,2]$, with $\int w(t)\,\mathrm{d}t/t = 1$, and $Y \geq 1$. Integrating (1) against $w(y/Y)\,\mathrm{d}y/y$ and substituting $y \mapsto m'/y$ in the second sum,

$$\lambda(m) = \int_0^\infty \Big[w\Big(\frac{y}{Y}\Big) + \chi(m')\,w\Big(\frac{m'}{yY}\Big)\Big]\sum_{\substack{b\mid m'\\ b<y}}\chi(b)\,\frac{\mathrm{d}y}{y}, \tag{2}$$

which is [8, (2.3)] with the factor $\chi(m')$ added. Take $Y = \sqrt{N}$. For $m = N - p \leq N$ the integrand vanishes unless $y \leq 2\sqrt{N}$, so all moduli b are below $2\sqrt{N}$.

2.3. The first half.

Summing the term $w(y/Y)$ of (2) over $p < N$ gives

$$\int_0^\infty w\Big(\frac{y}{Y}\Big)\sum_{b<y}\chi(b)\,\pi(N;b,N)\,\frac{\mathrm{d}y}{y}$$

(for b with $(b,N) > 1$ the progression contains at most one prime, and these $b \leq 2\sqrt{N}$ contribute $O(\sqrt{N})$). By H(N) with $X = N$,

$$\sum_{b<2\sqrt{N}}\Big|\pi(N;b,N) - \frac{\pi(N)}{\varphi(b)}\Big| \ll N\Big(\frac{\log\log N}{\log N}\Big)^2, \qquad \sum_{\substack{b<y\\ (b,N)=1}}\frac{\chi(b)}{\varphi(b)} = L(1,\chi)\mathfrak{E}_D(N) + O(y^{-1+\varepsilon}),$$

the second by Proposition 2.1 and the convergence of $\sum\chi(b)/\varphi(b)$ (partial summation against $\sum_{b\leq t}\chi(b) \ll D$). With $\int w(y/Y)\,\mathrm{d}y/y = 1$, the first half is

$$L(1,\chi)\mathfrak{E}_D(N)\,\pi(N) + O\Big(N\Big(\frac{\log\log N}{\log N}\Big)^2\Big).$$

2.4. **The second half.** The term $\chi(m')w(m'/(yY))$ summed over $p<N$ with $b\mid m'$, i.e. $b\mid N-p$ (as $(b,D)=1$), is

$$T_b(y)=\sum_{k\ge 0}\sum_{\substack{p<N,\ p\equiv N\ (b)\\ D^k\,\|\,N-p}}\chi\Big(\frac{N-p}{D^k}\Big)\,w\Big(\frac{N-p}{D^kyY}\Big).$$

If $D\mid N$, only $p=D$ has $D\mid N-p$, and the terms $k\ge 1$ are $O(1)$. Let $D\nmid N$. The inner sum is over the primes in the classes $p\equiv N\pmod b$, $p\equiv N+D^ka\pmod{D^{k+1}}$, $a\not\equiv 0\pmod D$, with the smooth weight $f_{k,y}(p)=w((N-p)/(D^kyY))$ and the value $\chi(-a)$; by the Chinese remainder theorem these are $\varphi(D)$ classes modulo bD^{k+1} (for $k=0$) or $D-1$ classes (for $k\ge 1$, a ranging over $1,\dots,D-1$), all coprime to bD when $(b,N)=1$. The smooth-weighted count of primes in a class modulo $q=bD^{k+1}$ is $\frac{1}{\varphi(q)}\sum_p f_{k,y}(p)$ plus an error which, by partial summation in f as in [8, Lemma 3.1], is bounded by $\sup_{X\le N}|\pi(X;q,\cdot)-\pi(X)/\varphi(q)|$ times $O(1)$; summed over $b<2\sqrt N$ this is the hypothesis $\mathrm{H}(N)$ for the moduli $q=bD^{k+1}\le 2\sqrt N D^{k+1}$, which lie below $\sqrt N(\log N)^A$ as long as $D^{k+1}\le(\log N)^{A-1}$. For the larger k, $D^k>(\log N)^{A-2}$, the trivial bound $\sum_b\#\{p:\ bD^k\mid N-p\}\ll\sum_{b<2\sqrt N}(N/(bD^k)+1)\ll N\log N/D^k+\sqrt N$ is $O(N/(\log N)^{A-3})$. The main terms: for $k=0$ the classes carry $\chi(N-a)$, a coprime to D, and $\sum_{(a,D)=1}\chi(N-a)/\varphi(bD)=c_D(N)/\varphi(b)$; for $k\ge 1$ the classes carry $\chi(-a)$, $a=1,\dots,D-1$, and $\sum_{a=1}^{D-1}\chi(-a)=0$. Hence

$$T_b(y)=\frac{c_D(N)}{\varphi(b)}\sum_{p<N}w\Big(\frac{N-p}{yY}\Big)+E_b(y),\qquad\sum_{b<2\sqrt N}\sup_y|E_b(y)|\ll N\Big(\frac{\log\log N}{\log N}\Big)^2.$$

Summing $\chi(b)T_b(y)$ over $b<y$ and integrating:

$$\int_0^\infty\sum_{p<N}w\Big(\frac{N-p}{yY}\Big)\frac{\mathrm dy}{y}=\sum_{p<N}\int_0^\infty w(t)\,\frac{\mathrm dt}{t}=\pi(N)$$

(substitute $t=(N-p)/(yY)$; $N-p\ge 1$), the b-sum is again $L(1,\chi)\mathfrak E_D(N)+O(y^{-1+\varepsilon})$ with $y\gg\sqrt N\,(N-p)/N$, and the range of y where $w((N-p)/(yY))\ne 0$ with $N-p\le N^{1-\varepsilon}$ (small y, where the truncation error $y^{-1+\varepsilon}$ is not small) contains $O(N^{1-\varepsilon})$ primes. The second half is therefore $c_D(N)L(1,\chi)\mathfrak E_D(N)\pi(N)+O(N(\log\log N/\log N)^2)$.

2.5. **Conclusion.** Adding the two halves and multiplying by w_D, $S_D(N)=w_DL(1,\chi)(1+c_D(N))\mathfrak E_D(N)\pi(N)+O(N(\log\log N/\log N)^2)$, and $w_DL(1,\chi_{-D})=2\pi/\sqrt D$ by the class number formula $L(1,\chi_{-D})=2\pi h/(w_D\sqrt D)$ with $h=1$. This is (i). Part (ii) is Linnik's theorem [14, Chapter 7], whose main term is $\pi A_0\prod_{\ell\mid N}(\dots)\,N/\log N$ with the same local factors; the identification of his constant with $\pi\mathfrak S_4(N)$ is the computation of Proposition 2.1 with $c_4=0$. □

Remark 2.2. Where $\mathrm{H}(N)$ enters. The Bombieri–Vinogradov theorem covers $b\le\sqrt N(\log N)^{-B}$ with the saving $(\log N)^{-A}$ and no dependence on the residue class. The window $\sqrt N(\log N)^{-B}<b<2\sqrt N$ contributes trivially $\ll\sum_b\pi(N;b,N)\ll\pi(N)\sum_b 1/\varphi(b)\asymp\pi(N)\,B\log\log N$, which is larger than the main term, and there is no cancellation to use in it, because the error terms $\pi(X;b,N)-\pi(X)/\varphi(b)$ enter with absolute values. This window is exactly what [1] controls for a fixed class, and what the dispersion method controls for the class N in the case $D=4$.

3. Numerical verification

The script `linnik_heegner.py` computes $S_D(N)$ exactly (the table of r_D by the divisor sum, the primes by a sieve), $\mathfrak S_D(N)$ by the Euler product over $\ell\le 2\cdot 10^6$ (tail $O(10^{-6})$), and the counts of primes $p<N$ with $N-p$ a norm and with $N-p$ a norm with $\Omega(N-p)\le 2$, for

eleven values $10^4 \le N \le 2\cdot 10^5$ (even, odd, prime, 3^9, 2^{16}, $60060 = 2^2\cdot 3\cdot 5\cdot 7\cdot 11\cdot 13$) and all nine D. Table 1 gives the extremes.

D	$2\pi/\sqrt{D}$	$1+c_D(N)$	$S_D(N)$	main term	ratio	ratio, 11 values of N	norm/π	norm, $\Omega\le 2$, /π
3	3.6276	1.5	179946	179317.7	1.0035	[0.9975, 1.0192]	0.500	0.340
4	3.1416	1	36700	36639.8	1.0016	[0.9798, 1.0233]	0.178	0.126
7	2.3748	7/6	24814	24870.0	0.9977	[0.9905, 1.0132]	0.244	0.166
8	2.2214	1	55494	54859.3	1.0116	[0.9624, 1.0139]	0.301	0.187
11	1.8945	0.9	41198	41450.8	0.9939	[0.9741, 1.0195]	0.259	0.164
19	1.4415	17/18	25004	25181.6	0.9929	[0.9646, 1.0394]	0.209	0.138
43	0.9582	43/42	26338	26328.1	1.0004	[0.9516, 1.0170]	0.223	0.148
67	0.7676	67/66	20364	20386.9	0.9989	[0.9505, 1.0232]	0.195	0.136
163	0.4921	163/162	12798	12705.7	1.0073	[0.9724, 1.0406]	0.149	0.109

TABLE 1. $S_D(N) = \sum_{p<N} r_D(N-p)$ against $\frac{2\pi}{\sqrt{D}}\mathfrak{S}_D(N)\pi(N)$ at $N = 2\cdot 10^5$ ($\pi(N) = 17984$), the range of the ratio over the eleven N of the script, and the proportions of primes $p < 2\cdot 10^5$ with $N-p$ a norm, and a norm with at most two prime factors. The ratios deviate most at $N \approx 10^4$ and for the larger D, where $\mathfrak{S}_D(N)\pi(N)$ is smallest. Script `linnik_heegner.py`, file `results/linnik_heegner.txt`; exact integer counts, floating point in the Euler products.

Three features of the data. (a) The factor $1+c_D(N)$ is real: for $D=3$ the ratios without it are 0.508, 1.499, 1.008, 1.501, 1.496, 1.006, 0.509, 0.502, 0.510, 1.002, 1.505 for the eleven N, i.e. $\frac12, \frac32, 1$ according to $N \equiv 1, 2, 0 \pmod 3$; for $D=7$ they are 0.836 and 1.156, i.e. $1 \mp \frac16$; for $D = 4$ and 8 they are already 1. (b) At odd N and $D \equiv 3 \pmod 8$ the count of p with $N-p$ a norm with $\Omega \le 2$ is 0 or 1 (the parity condition of Section 1); at even N it is between $0.11\,\pi(N)$ ($D = 163$) and $0.34\,\pi(N)$ ($D = 3$), against 0.13–0.50 for norms with any number of prime factors. (c) The sieve densities of $\{N-p\}$ at $N = 2\cdot 10^5$, $D = 4$: the proportion of p with $(N-p, P(z)) = 1$ is 0.4160, 0.2796, 0.2112 for $z = 10, 30, 100$ against $\prod_{q<z,\, q\nmid N}(1-\frac{1}{q-1}) = 0.4167$, 0.2802, 0.2122 (dimension 1), and the proportion with no inert prime factor below y is 0.2910, 0.2412, 0.2100 for $y = 10^2, 10^3, 10^4$ against the half-dimensional product 0.2903, 0.2362, 0.2048; at $z = 10^3$ and $y = 10^5$ the sifting range reaches $\sqrt{N}$ and the products no longer describe the counts (1.098 and 0.926), which is the sieve limit, not a defect of the data.

4. What is proved and what is not

(1) The asymptotic formula for $S_D(N)$ with the explicit singular series, for all nine D, under Hypothesis H(N) (Theorem 1.2(i)); unconditionally for $D = 4$ (Linnik). The new content is the uniform treatment, the ramified factor $1+c_D(N)$, and the identification of the exact input, a level $\sqrt{N}(\log N)^A$ for the single residue class N.

(2) The sieve dimension of $\{N-p\}$ is 1 (trivial, and not the point); the dimension of the norm-weighted sequence is 1 on average with a half-dimensional inert part; its level of distribution is not known, and in particular it is not given by the Bombieri–Vinogradov theorem (Proposition 1.4(ii)). Consequently no statement "$N = p + P_k$ with $N-p$ a norm" is proved here for any k, and the lower bound that a Chen-type argument would produce, $f_1(5) - \frac12 \int F_1(\dots)$ in some form, has no meaning until the level is established; we have not written such a constant.

(3) What remains unconditional and new in the sieve direction is the shape of the problem: one obstacle of Parts I–IV, the dimension 2, is gone; the other, the fixed N, is not, and it now sits in the residue class rather than in the Kloosterman argument. The natural

next steps are (a) the dispersion method for N_D, $D \neq 4$, which would make Theorem 1.2 unconditional, and (b) the half-dimensional sieve of Iwaniec [11, 12] for the inert primes, combined with the linear sieve for the small primes, applied to the unweighted $\{N - p\}$ at the level $1/2$: it would give $N = p + m$ with m free of small primes and of inert primes below $N^{1/2-\varepsilon}$, hence a norm up to at most one inert prime factor above $N^{1/2-\varepsilon}$, and the switching needed to remove that factor is a problem of the Chen type with the half-dimensional densities, whose constants we have not computed.

Data and code

The script `linnik_heegner.py` (Section 3; Python 3.12, numpy; about two minutes; exact integer counts) with its output `results/linnik_heegner.txt` is in the directory `two-squares-part5` alongside Parts I–IV. All computations are evidence, not proof; the theorems do not depend on them.

References

[1] E. Bombieri, J. B. Friedlander and H. Iwaniec, Primes in arithmetic progressions to large moduli. III, J. Amer. Math. Soc. 2 (1989), 215–224.

[2] B. M. Bredihin, Binary additive problems of indeterminate type. II. Analogue of the problem of Hardy and Littlewood, Izv. Akad. Nauk SSSR Ser. Mat. 27 (1963), 577–612.

[3] J. R. Chen, On the representation of a larger even integer as the sum of a prime and the product of at most two primes, Sci. Sinica 16 (1973), 157–176.

[4] R. Chen, Sums of two squares with few prime factors via theta series of quaternary forms, preprint, version 3 (2026), doi:10.5281/zenodo.23826176.

[5] R. Chen, Sums of two squares with few prime factors, II, preprint, version 3 (2026), doi:10.5281/zenodo.23866064.

[6] R. Chen, Sums of two squares with few prime factors, III, preprint (2026), doi:10.5281/zenodo.23826860.

[7] R. Chen, Sums of two squares and Heegner norms with few prime factors, IV, preprint (2026), doi:10.5281/zenodo.23866356.

[8] J. B. Friedlander and H. Iwaniec, On a theorem of Bredihin and Linnik, preprint (2018), arXiv:1807.06648.

[9] H. Halberstam and H.-E. Richert, Sieve Methods, Academic Press, London, 1974.

[10] C. Hooley, On the representation of a number as the sum of two squares and a prime, Acta Math. 97 (1957), 189–210.

[11] H. Iwaniec, Primes of the type $\varphi(x, y) + A$ where φ is a quadratic form, Acta Arith. 21 (1972), 203–234.

[12] H. Iwaniec, The half dimensional sieve, Acta Arith. 29 (1976), 69–95.

[13] J. D. Lichtman, Primes in arithmetic progressions to large moduli, and Goldbach beyond the square-root barrier, preprint (2023), arXiv:2309.08522.

[14] Yu. V. Linnik, The Dispersion Method in Binary Additive Problems, Translations of Mathematical Monographs 4, Amer. Math. Soc., Providence, 1963.

[15] J. Maynard, Primes in arithmetic progressions to large moduli III: Uniform residue classes, preprint (2020), arXiv:2006.08250.

[16] H.-E. Richert, Selberg's sieve with weights, Mathematika 16 (1969), 1–22.

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